

Low Speed Aerodynamics (LSA)

8-Week Mentorship Plan

I recommend that you learn simulation software like ansys, if you want to get hands on experience on Aerodynamics. Each week provides a clear primary goal, core topics, hands-on activities, and reference materials. Try to complete the Mentee Activity so that you'll get your concept clear, Don't take it for the last moment You need to submit mid term report, end term report and Video explaining your project document(Use overleaf or docs).

Week 1: Governing Equations & Special Flow Conditions

Weekly Goal

Master the fundamental equations of fluid motion, understand the core simplifications (2D, steady, incompressible), and derive classical viscous flow solutions.

Key Topics	• Fundamental Flows: Poiseuille Flow (pressure-driven) & Couette Flow • Navier-Stokes Simplifications • Introduction to Boundary Layers & the Blasius solution
Mentee Activity	Derive the velocity profile for Poiseuille flow from the generalized mass conservation equations.
Useful Links	Detailed Blasius Solution Derivation (PDF)

Week 2: Boundary Layer Dynamics & Separation

Weekly Goal

Comprehend the behavior of fluids near solid boundaries, differentiate between laminar and turbulent layers, and understand the physical mechanisms of flow separation.

Key Topics	• Laminar vs. Turbulent Boundary Layers (Velocity gradients) • Skin Friction coefficients across varying Reynolds numbers • Flow Separation, separation bubbles, and turbulent boundary layer advantages
Mentee Activity	Sketch and compare the velocity profiles of a laminar versus a turbulent boundary layer, highlighting maximum skin friction points.
Useful Links	Aviation StackExchange: Flow Separation & Stall Mechanisms

Week 3: Inviscid Flows & Potential Flow Fundamentals

Weekly Goal

Learn to construct complex flow fields around objects by superimposing basic, elementary inviscid flows (sources, sinks, dipoles, and uniform flow).

Key Topics	• Flow Visualization: Differentiating streamlines, pathlines, and streaklines • Basic Elementary Flows: Uniform flow, dipoles, and vortices • Flow Around a Cylinder (Combining flows for rotating/non-rotating models)
Mentee Activity	Use the interactive simulator to combine a uniform flow and a dipole, visualizing the flow field over a cylinder.
Useful Links	Interactive Web Tool: Potential Flow Simulator

Week 4: Vortex Dynamics & The Generation of Lift

Weekly Goal

Understand the mathematical and physical nature of vortices and apply the Biot-Savart law and Kutta-Joukowski theorem to calculate aerodynamic lift.

Key Topics	• Free (irrotational) vs. Forced (solid body) vortices • Vortex Core Models (Rankine, Scully, Lamb-Oseen) • Biot-Savart Law & the Kutta-Joukowski Theorem
Mentee Activity	Calculate the theoretical lift per unit span of a spinning cylinder using the Kutta-Joukowski theorem.
Useful Links	YouTube: Solid Body Rotation vs Irrotational Vortex Reference: Plane Potential Flows

Week 5: Inviscid Airfoil Modeling (Thin Airfoil Theory)

Weekly Goal

Apply Thin Airfoil Theory and utilize the Kutta Condition to mathematically model and predict the aerodynamic characteristics of 2D asymmetric airfoils.

Key Topics	• The Kutta Condition (Enforcing smooth flow leaving the trailing edge) • Vortex Sheets & Bound Circulation • Thin Airfoil Derivation using Fourier series
Mentee Activity	Write a brief summary explaining why the Kutta condition is the vital physical bridge required for inviscid mathematical models to work.
Useful Links	<i>Search:</i> “Thin Airfoil Theory Derivation” or “The Kutta Condition Aerodynamics”

Week 6: Airfoil Geometry & Wing Parameters

Weekly Goal

Translate aerodynamic theory into practical design by analyzing how physical geometry (camber, thickness) and high-lift devices alter wing performance.

Key Topics	• Geometric Parameters: Chord, mean camber line, maximum thickness • Effects of thickness and camber on lift coefficient curves and stall • High Lift Devices: Flaps (split, slotted, Fowler) and leading-edge Slats
Mentee Activity	Compare the lift curves of a symmetrical airfoil versus a highly cambered airfoil, noting the shifts in the zero-lift angle of attack.
Useful Links	AeroToolbox: Aircraft Flaps and High Lift Devices

Week 7: 3D Aerodynamics & Lifting Line Theory

Weekly Goal

Transition from 2D infinite airfoils to 3D finite wings by understanding wingtip vortices, downwash, and the creation of induced drag.

Key Topics	• Finite Wing Effects: Air spillage from high to low pressure zones • 3D Vortex Systems: Bound, Trailing, and Starting vortices • Downwash & Induced Drag formulation
Mentee Activity	Mathematically and conceptually explain how a wing's aspect ratio acts as the primary driver for the magnitude of induced drag.
Useful Links	<i>Search:</i> "Prandtl's Lifting Line Theory" or "Wingtip Vortices and Induced Drag"

Week 8: Computational Approaches (Panel Methods)

Weekly Goal

Grasp the fundamentals of Panel Methods and the Vortex Lattice Method (VLM) to solve complex aerodynamic geometries where Lifting Line Theory fails.

Key Topics	• Limitations of Lifting Line Theory (Sweep, low aspect ratio, complex shapes) • Panel Method Basics (Discretizing surfaces, distributing singularities) • Applying Boundary Conditions & the Vortex Lattice Method (VLM)
Mentee Activity	Outline the foundational computational steps required to set up a basic Vortex Lattice Method simulation for a swept wing.
Useful Links	<i>Search:</i> "Vortex Lattice Method (VLM) Explained" or "Aerodynamic Panel Methods"